

# Paper II — The Photon as a Bound Pair of Spin- $\frac{1}{2}$ Lattice Quanta: A Non-Birefringent, Luminal, Transverse Composite

**B. Ludwig** *Independent researcher*

**Notebook source material:** M. Ludwig (2007), *Physics Notes*, pp. 5–6 (“spinor electrodynamics”)

*Series: “A Universe in a Bottle” — Paper II of XI. Builds directly on Paper I (BCC QCA substrate; speed of light as rotation rate).*

*Revision 2 (2026-06-08): re-issued for the eleven-paper series; cell-size statements updated to the canonical F107 value  $a = 6.5978 \ell_P$  adopted in Paper VII.*

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## Abstract

We construct the electromagnetic photon not as a fundamental spin-1 gauge boson but as a bound pair of two spin- $\frac{1}{2}$  Weyl quanta of the underlying body-centred-cubic (BCC) quantum cellular automaton — the lattice realisation of de Broglie’s neutrino theory of light. The pair carries total momentum  $\mathbf{k}$  shared as  $\mathbf{k}/2$  between two constituents on opposite chiral branches, so its real  $(\mathbf{E}, \mathbf{B})$  field rotates at the **helicity-symmetric** rate  $\Omega_{\text{pair}}(\mathbf{k}) = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) \equiv \Omega_{\text{even}}(\mathbf{k})$ . We show this object is (i) massless and luminal, with group velocity  $\rightarrow 1/\sqrt{3} = c_{\text{lat}}$ ; (ii) transverse and spin-1, with two physical polarisations; and (iii) **exactly non-birefringent**, because both helicities ride the single rate  $\Omega_{\text{pair}}$  — there is no single-branch photon to split against. This last property is decisive: an earlier composite construction (the  $\sigma$ -bilinear photon) assigned each helicity to its own branch and was therefore birefringent at first order in  $k$ , a prediction excluded by gamma-ray-burst (GRB) and active-galactic-nucleus (AGN) polarimetry by many orders of magnitude at the gravitationally-fixed cell size. The paired photon removes that prediction entirely and is independently forced from three directions — observation,  $U(1)$  minimal coupling, and Ludwig’s notebook construction — that all land on the same dispersion. All claims are verified to machine precision (5/5 checks; residuals 0 to  $\sim 10^{-15}$ ).

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## 1. Introduction

Paper I established that on the BCC QCA a correlated pair of lattice spinors furnishes a real field pair  $(\mathbf{E}, \mathbf{B})$  whose free evolution is an exact rotation, and that the speed of light is the rotation rate of that pair per unit wavenumber. The present paper answers the next question: *what, precisely, is the photon?*

In the Standard Model the photon is a fundamental spin-1 gauge field  $A_\mu$  introduced by hand to gauge  $U(1)$ . On the lattice no such field need be posited. Following a proposal recorded in Ludwig’s 2007 notebook (pp. 5–6) — and ultimately de Broglie’s

1930s neutrino theory of light — the photon is a **bound pair of two spin- $\frac{1}{2}$  “fermion-photons”  $\gamma_{1/2}$  that behave together as a single boson and only occur as a pair; they do not occur separately.** This is structurally the same idea as a Cooper pair in superconductivity: two spin- $\frac{1}{2}$  objects bound into a single bosonic excitation.

The composite photon arises *natively* from the lattice’s own Weyl quanta and requires no additional field construct, which is its chief economy over the Standard-Model photon. The task of this paper is to make the construction precise, derive its propagation law, and confront its observational consequences.

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## 2. The two-branch structure of the BCC walk

From Paper I (Eq. 2.1) the free Weyl walk has two dispersion branches,

$$\omega^\pm(\mathbf{k}) = \arccos(c_x c_y c_z \pm s_x s_y s_z), \quad c_i = \cos \frac{k_i}{\sqrt{3}}, \quad s_i = \sin \frac{k_i}{\sqrt{3}},$$

related by parity  $\omega^+(-\mathbf{k}) = \omega^-(\mathbf{k})$ . The two branches are the two chiralities/helicities of the walk. A real electromagnetic field is built from the Riemann-Silberstein (RS) combinations

$$\mathbf{F}_\pm(\mathbf{k}) = \mathbf{E}(\mathbf{k}) \pm i\mathbf{B}(\mathbf{k}),$$

which are exactly the eigenvectors of the one-tick rotation matrix  $R(\Omega)$  of Paper I:  $R(\Omega)(1, \mp i)^\top = e^{\mp i\Omega}(1, \mp i)^\top$ . Reality of  $(\mathbf{E}, \mathbf{B})$  forces  $\mathbf{F}_+(-\mathbf{k}) = \mathbf{F}_-^*(\mathbf{k})$ .

### 2.1 The fork: two ways to assign helicities to branches

There are two consistent ways to propagate the RS eigenstates while preserving the reality (Hermitian-symmetry) constraint:

- **Chiral assignment (the  $\sigma$ -bilinear photon, F29/F37/F39):** let each helicity ride its own branch,  $\mathbf{F}_+ \rightarrow \Omega^+ = 2\omega^+(\mathbf{k}/2)$  and  $\mathbf{F}_- \rightarrow \Omega^- = 2\omega^-(\mathbf{k}/2)$ . Hermitian symmetry is preserved because  $\Omega^+(-\mathbf{k}) = \Omega^-(\mathbf{k})$ .
- **Even assignment (the paired photon, this paper):** let *both* helicities ride the single symmetric rate  $\Omega_{\text{pair}} = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) = \Omega_{\text{even}}$ .

These two laws are physically inequivalent, and §4 shows the first is observationally dead while the second survives. The chiral law is retained only for the massive/confined sectors (Papers V-VI), which carry no astrophysical vacuum-birefringence bound. The systematic statement of which sector rides which law is the pairing classification theorem of Paper IV (Finding F91): the propagator law is *forced* by the chirality structure of each coupling, not chosen.

### 3. The paired photon

#### 3.1 Construction

The photon is a bound  $(+, -)$  pair: total momentum  $\mathbf{k}$  shared by two Weyl constituents at  $\mathbf{k}/2$  each, one on the  $+$  branch and one on the  $-$  branch. Because the pair is bound and symmetric, the phase it accumulates per tick is the **sum** of the constituent phases:

$$\boxed{\Omega_{\text{pair}}(\mathbf{k}) = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) \equiv \Omega_{\text{even}}(\mathbf{k}).}$$

This is exactly the even-law rotation rate of the lattice's  $(\mathbf{E}, \mathbf{B})$  propagator from Paper I — no new propagator is invented; the pairing *names* the even law as the photon law and supplies its physical interpretation. The free field then evolves by the exact rotation (Paper I, Eq. 3.1) with  $\Omega = \Omega_{\text{pair}}$ . The propagator is implemented as `ca_photon_pair.py` (propagator = even law `ca_wmu._f26_rotation_step`).

#### 3.2 Why non-birefringent — the mechanism

In the chiral construction the two photon helicities  $\mathbf{F}^\pm$  were assigned to the two branches independently, so a generic linear polarisation (a superposition of  $\mathbf{F}^+$  and  $\mathbf{F}^-$ ) acquired a relative phase  $\Delta\Omega N$  after  $N$  ticks and its plane of polarisation rotated — vacuum birefringence. In the **paired** photon there is no single-branch photon: it “only occurs as a pair,” always containing both branches symmetrically, so both helicities of the resulting field ride the *one* rate  $\Omega_{\text{pair}}$ . The split has nothing to split against. Formally, the birefringence

$$\Delta\Omega(\mathbf{k}) = \Omega^+(\mathbf{k}) - \Omega^-(\mathbf{k})$$

which is non-zero for the chiral law (see §4), never enters the paired-photon evolution because the pair is built symmetrically in  $\pm$  from the start.

#### 3.3 Massless, luminal, transverse, spin-1

Direct evaluation of (3.1) gives:

- **Massless and luminal:**  $\Omega_{\text{pair}}(0) = 0$  and  $\Omega_{\text{pair}}/|\mathbf{k}| \rightarrow 0.577350 = 1/\sqrt{3}$ , with group velocity  $d\Omega_{\text{pair}}/d|\mathbf{k}| \rightarrow 1/\sqrt{3} = c_{\text{lat}}$  (verified to  $7.7 \times 10^{-6}$  by finite difference; the slope along the body diagonal evaluates to 0.5773503).
- **Transverse, spin-1:** the field has two transverse polarisations,  $|\mathbf{E} \cdot \hat{\mathbf{k}}|/|\mathbf{E}| = 1.8 \times 10^{-17}$ ; it is real ( $10^{-15}$ ) and norm-conserving ( $6 \times 10^{-15}$ ).
- **“Only as a pair”:** the pair phase equals the constituent sum  $\omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2)$  exactly (residual 0); an unpaired single branch would carry  $\Omega^\pm$ , which differs (median split  $5.3 \times 10^{-3}$ ).

The two transverse polarisations and the masslessness together give the photon its spin-1 content without any spin-1 field being fundamental: the spin-1 object is the

symmetric combination of two spin- $\frac{1}{2}$  constituents ( $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$ , with the antisymmetric singlet being the pseudoscalar partner — the pion-channel sibling of Paper XI).

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#### 4. Confrontation with polarimetry: why the chiral photon is excluded

Along the BCC body diagonal  $(1, 1, 1)$  the chiral-law birefringence is, to leading order (expansion of (2.1)),

$$\Delta\Omega \approx -\frac{\sqrt{3}}{27} k^2, \quad \frac{\Delta v_\phi}{c_{\text{lat}}} \approx -\frac{k}{18},$$

i.e. **linear vacuum birefringence** in photon energy ( $k \propto E_\gamma/E_{\text{Planck}}$ ); it vanishes on the cube axes and is maximal along the body diagonals. Linear birefringence rotates the polarisation angle of light from a distant source by  $\Delta\theta \sim \Delta k \cdot d$ , and is tightly constrained by GRB and AGN polarimetry (INTEGRAL, *Fermi*/GBM). At the cell size that the gravity sector independently requires (Paper VII;  $a = 6.5978 \ell_P = 1.07 \times 10^{-34}$  m), the chiral law overshoots the polarimetry bound by many orders of magnitude — it is excluded precisely at the gravitationally-fixed cell size (Findings F65/F66).

The paired photon (3.1) removes this prediction entirely: there is no helicity split, so a linearly polarised body-diagonal mode shows the symmetric phase  $S = \phi_+ + \phi_- = -1.8 \times 10^{-15}$  (exactly non-birefringent) where the chiral law gives  $S = +0.0792 = -\Delta\Omega N$ . This is the decisive empirical reason the paired photon supersedes the  $\sigma$ -bilinear as *the* electromagnetic photon (Finding F67).

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#### 5. Three independent routes to the same object

The paired photon is not an *ad hoc* repair. It is forced from three independent directions that now agree:

1. **Observation (§4):** the photon must be non-birefringent  $\Rightarrow$  it must ride the helicity-symmetric rate  $\Omega_{\text{even}}$ .
2.  **$U(1)$  minimal coupling (Paper IV; Finding F68):** the charge-coupling photon is the  $U(1)$  *identity channel*  $e^{i\theta}\mathbf{I}$ , which acts identically on both helicities — helicity-blind, hence  $\Omega_{\text{even}}$ . The general statement (F91) is that the unique identity-channel direction of any neutral coupling  $aT_3 + bY/2$  is the one proportional to electric charge  $Q$ ; that direction *is* the photon.
3. **Ludwig’s notebook (pp. 5-6):** the photon is a bound pair of two  $\gamma_{1/2}$ , symmetric in  $(+, -)$ , hence rate  $\omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) = \Omega_{\text{even}}$ .

All three land on the same dispersion. The notebook’s open question “Is there a scalar photon?” (p. 6) is thereby answered: the scalar/identity photon **is** the paired photon, and it is the physical one.

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## 6. Consistency with the gravity sector

The non-birefringence of the paired photon does more than settle a polarimetry tension: it **restores the cell size the gravity sector requires** (Paper VII). The structural (Sakharov-induced) value of Newton’s constant pins the cell at the parameter-free value  $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P$  (Finding F79/F107). The chiral photon was falsified *at that very cell size*; the paired photon survives all astrophysical vacuum-birefringence bounds there, with the Lorentz-violation ( $n = 2$ ) cutoff at  $E_{\text{QG},2} = \sqrt{54} \hbar c / a \approx 1.36 \times 10^{19} \text{ GeV}$  — above the current GRB/LHAASO time-of-flight bounds. The model’s dimensionless predictions (Weinberg angle, mass ratios,  $c_{\text{lat}} = 1/\sqrt{3}$ , PPN  $\beta = \gamma = 1$ , Mercury precession, factor-2 light bending) are cell-size-independent and unchanged; what the paired photon buys is that the gravity-fixed  $a$  is no longer ruled out by the photon sector. This co-certification ties Papers II, VII and VIII together.

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## 7. Verification summary

| #   | Check                                                                                                       | Result                                                             | Status       |
|-----|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------|
| PP1 | $\Omega_{\text{pair}} = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) =$<br>even-law rate<br>(2000 modes) | residual 0                                                         | exact        |
| PP2 | Non-birefringent:<br>linearly polarised<br>body-diagonal<br>mode, 6 ticks                                   | paired<br>$S = -1.8 \times 10^{-15}$<br>vs chiral<br>$S = +0.0792$ | machine      |
| PP3 | Massless &<br>luminal:<br>$\Omega_{\text{pair}}/k \rightarrow$<br>0.57735,<br>$v_g \rightarrow 1/\sqrt{3}$  | err $7.7 \times 10^{-6}$                                           | quantitative |
| PP4 | “Only as a pair”<br>(constituent sum<br>vs single-branch<br>split)                                          | residual 0                                                         | machine      |
| PP5 | Spin-1, two<br>transverse<br>polarisations, real,<br>norm-conserving                                        | $\leq 6 \times 10^{-15}$                                           | machine      |

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Underlying findings: F69 (paired-spinor photon), F67/F68 (even-law vs bilinear; minimal coupling forces the even photon), F65/F66 (helicity-chirality map; all-sky bire-

fringe bound), F37 (RS/BCC chirality), F39 (the retired two-helicity bilinear), F91 (pairing classification theorem).

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## 8. Discussion

**What is gained.** The photon is now an emergent composite of the lattice’s own quanta, with no new fundamental field. Its masslessness, luminality, transversality and — crucially — its exact non-birefringence are all consequences of the symmetric pairing, not inputs. The construction unifies an observational constraint, a gauge-theoretic requirement, and a 2007 notebook conjecture.

**What is retained for other sectors.** The chiral  $\sigma$ -bilinear is *not* discarded wholesale: it remains the correct *field construction* for the massive and non-Abelian sectors —  $W$ ,  $Z$ , and gluons (Papers V, VI) — which are massive or confined and carry no astrophysical vacuum-birefringence bound. Indeed the pairing classification theorem (F91) shows the propagator *law* (even vs chiral) is forced sector by sector:  $\gamma$  even (forced),  $W^\pm$  chiral (forced; left-projector coupling),  $Z$  even for its vector part with a mass-suppressed axial split, gluon even (forced; colour coupling is branch-blind).

**Open items.** (i) A genuine two-constituent bound-state simulation of the pair — the “negative binding energy” of the notebook and the massless-pair condition — beyond the present dispersion-level treatment (partly addressed by the bound-state machinery of Findings F73/F74 used in Paper XI). (ii) Whether a clean paired construction exists for the massive sectors.

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## References

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2. A. Bisio, G. M. D’Ariano, P. Perinotti, A. Tosini, “Weyl, Dirac and Maxwell quantum cellular automata,” (2015).
3. I. Bialynicki-Birula, “Photon wave function,” *Prog. Opt.* **36**, 245 (1996) — Riemann–Silberstein vector.
4. P. Laurent *et al.*, “Constraints on Lorentz Invariance Violation using INTEGRAL/IBIS,” *Phys. Rev. D* **83**, 121301 (2011); *Fermi*/GBM and INTEGRAL gamma-ray polarimetry constraints on vacuum birefringence.
5. M. Ludwig, *Physics Notes* (2007), pp. 5–6 (“spinor electrodynamics; two  $\gamma_{1/2}$  that only occur as a pair”).
6. Project findings: F69 (paired-spinor photon), F68 (minimal coupling forces the even photon), F67 (even-law vs bilinear mutually exclusive), F66/F65 (birefringence bounds; helicity-chirality map), F37 (RS/BCC chirality), F39 (two-helicity composite, retired as the photon), F91 (pairing classification).

*Companion papers: I (substrate), IV (electromagnetism), VII (gravity), VIII (black hole / cell-size co-certification).*